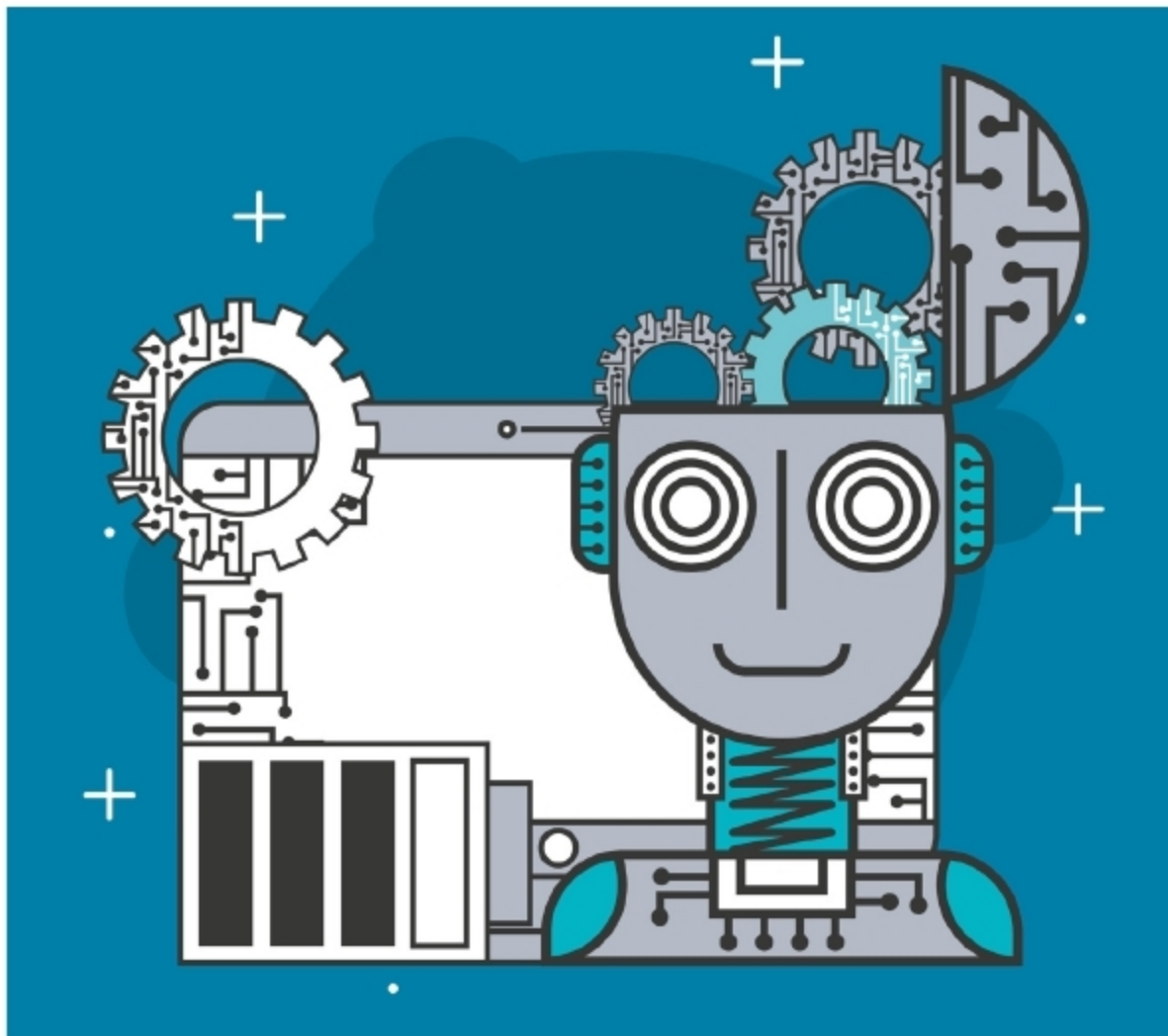


Using PyTorch with Google Colab for Machine Learning

PyTorch is a free and open source, deep learning library developed by Facebook. Google Colab is a free online cloud based tool that lets you deploy deep learning models remotely on CPUs and GPUs. This article is an introduction to PyTorch, and will demonstrate its benefits by using a linear regression model to predict the value of a given piece of land, based on its size.



Machine learning and data science have emerged as some of the most sought after fields in the tech community. The number of use cases of machine learning is growing continuously — from self-driving cars to smart home appliances. Open source deep learning libraries such as PyTorch and TensorFlow have been gaining a lot of steam. Within just three years of its initial release in 2016,

PyTorch has become very popular with companies like Apple, Verizon, NVIDIA and Uber, and is being used in their deep learning projects.

PyTorch

The initial purpose behind creating PyTorch was to replace the use of Numpy arrays with the more sophisticated tensors, which are similar, except that the latter perform much better with GPUs. It has now become

a market leader and competitor to Google's TensorFlow library.

The main component of PyTorch is the Torch library, written in the Lua scripting language, which provides the facility to create n-dimensional tensors (arrays) that support all the basic operations such as slicing, indexing and transposing. Additionally, it supports operations such as max, min and sum, and the more advanced tensor operations like vector multiplication. Development on the Torch library stopped in 2018. However, the PyTorch library is in active development by Facebook. The latest version of PyTorch was released in October 2019.

What is Google Colab?

The Google Colab (short for Colaboratory) is a research project launched by Google. Its intended purpose is to make the life of developers easier by giving them the facility to remotely deploy deep learning models on the CPUs and GPUs being offered in the form of virtual machines. It is based on the powerful open source technology, Jupyter, with which you can create an IPython notebook. The IPython notebook not only gives you the ability to write code, but to also tell a story through it, in an illustrated manner. You can execute code, create beautiful and vivid visualisations of your data, and write about each step you make in plain text or HTML. This makes your code